

CE725: Vibration-Based Structural Health Monitoring

End Semester Assignment – Part II Report

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1. Objective

The objective of this assignment was to identify an ARX model for a 5-storeyed structural system using measured input-output vibration data. The first 10 seconds of data were used for model identification, while the last 10 seconds were used for validation and prediction assessment.

The available signals were:

- Inputs: Forces at stories 2 and 5
- Outputs: Accelerations at stories 3 and 5

Thus, the system has:

$$r = 2 \quad \text{inputs}, \quad m = 2 \quad \text{outputs}$$

Sampling rate:

$$f_s = 100 \text{ Hz}$$

Total duration:

$$20 \text{ sec} = 2000 \text{ samples}$$

2. ARX Model Formulation

The ARX model of order p is written as:

$$Y = \theta V$$

where:

- Y : output matrix

- V : regressor matrix formed using present/past inputs and past outputs
- θ : unknown parameter matrix

The least squares estimate was obtained as:

$$\hat{\theta} = YV^T(VV^T)^{-1}$$

A reusable Python function was created in Jupyter Notebook to estimate the ARX model for any chosen order p .

3. Part (a): ARX Model of Order 4

An ARX model of order:

$$p = 4$$

was identified using the first 10 seconds of data.

The estimated parameter matrix had size:

$$\hat{\theta} \in \mathbb{R}^{2 \times 18}$$

3.1 Measured vs Reconstructed Responses

Measured outputs and reconstructed outputs over the first 10 seconds were compared.

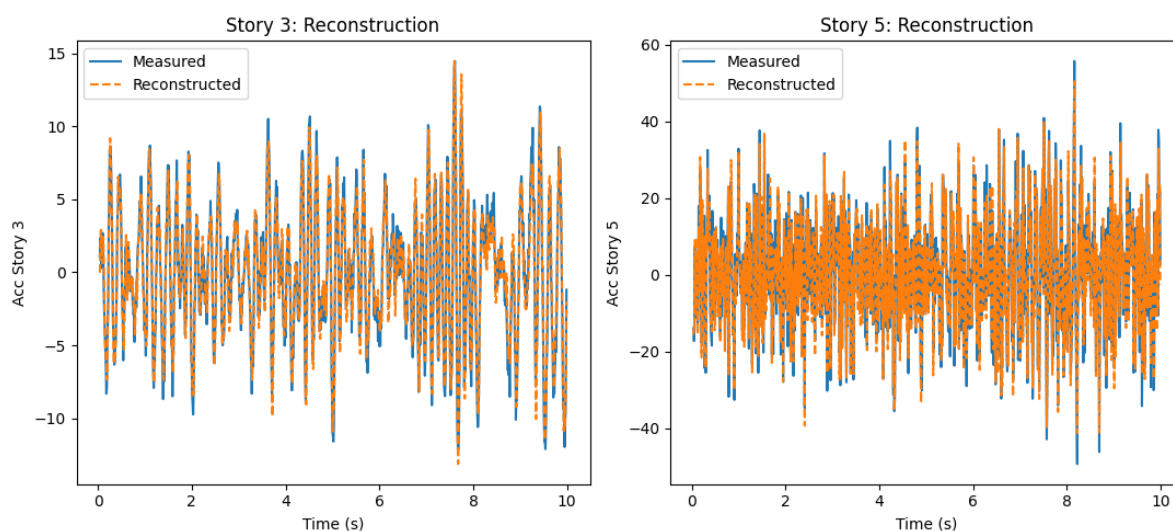


Figure 1: Measured and reconstructed responses for ARX(4) model using identification data.

The reconstructed responses closely matched the measured signals, indicating good fitting performance.

3.2 Measured vs Predicted Responses

The same identified model was used to predict responses for the last 10 seconds.

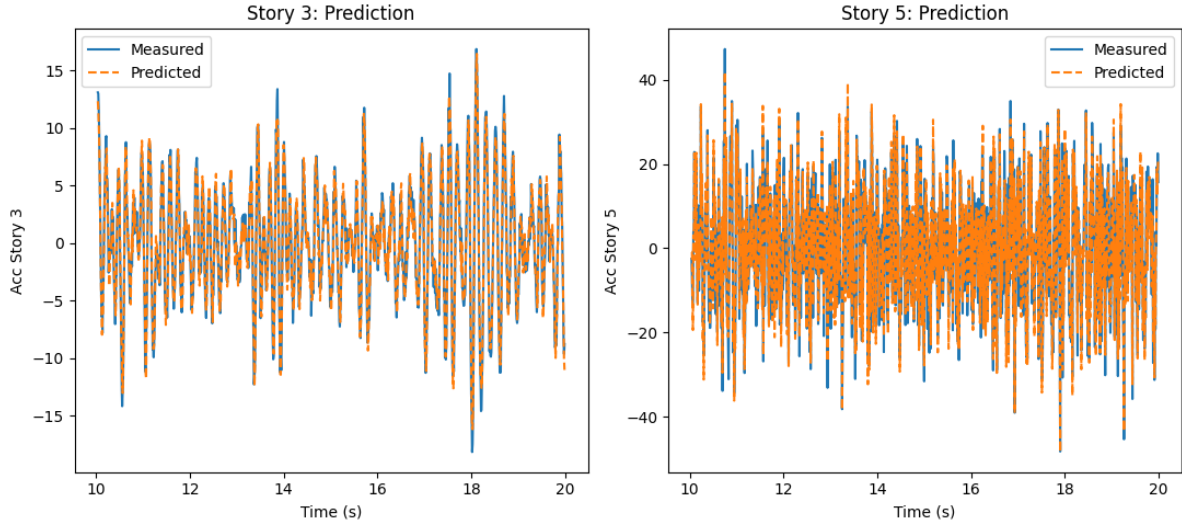


Figure 2: Measured and predicted responses for ARX(4) model using validation data.

The predicted responses followed the measured outputs reasonably well.

3.3 Error Ratios

Residual ratio:

$$\frac{RSS}{SSS} = 0.0647$$

Prediction ratio:

$$\frac{PESS}{PSSS} = 0.0744$$

Low values indicate that the model explained most of the signal energy successfully.

4. Part (b): AIC and BIC for Orders 4 to 103

ARX models were estimated for orders:

$$p = 4 \text{ to } 103$$

For each model, Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) were computed.

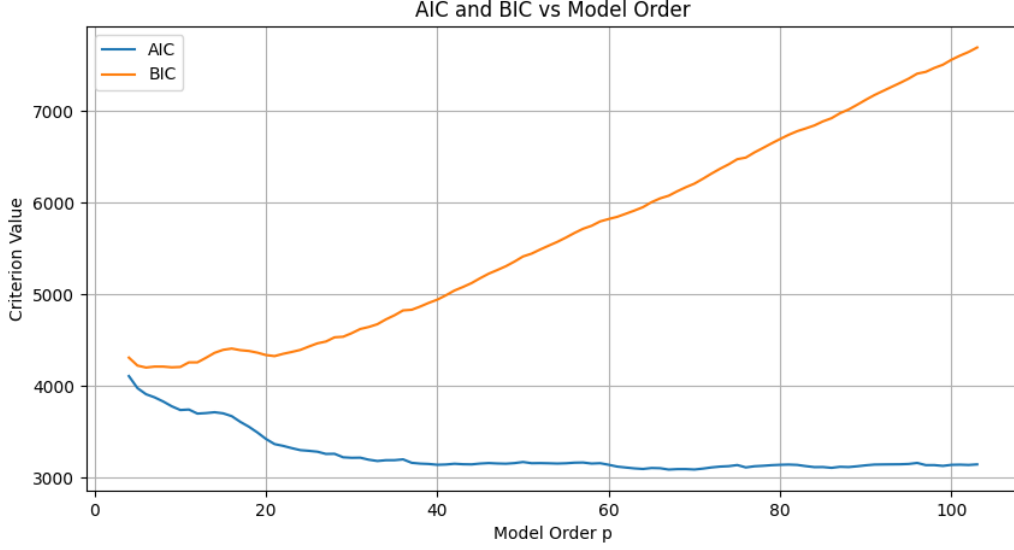


Figure 3: Variation of AIC and BIC with model order.

The minimum BIC was obtained at:

$$p = 6$$

Hence, ARX(6) was selected as the most parsimonious model.

5. Part (c): Error Ratios for Different Orders

For all model orders from 4 to 103, the following were computed:

- Residual ratio RSS/SSS
- Prediction ratio $PESS/PSSS$



Figure 4: Residual and prediction error ratios versus model order.

Observations:

- Residual ratio decreased continuously with increasing model order.
- Prediction ratio first decreased and then increased.
- Minimum prediction ratio occurred near:

$$p = 36$$

This demonstrates the bias-variance tradeoff. Lower-order models underfit, while excessively higher-order models overfit.

6. Part (d): Recursive Updating of ARX(6)

An initial ARX(6) model was estimated using the first 1 second of data (100 samples). Thereafter, the model was updated sample-by-sample up to 10 seconds using the recursive update equation:

$$\theta_{new} = \theta_{old} + (Y_{new} - \theta_{old}V_{new}) \frac{V_{new}^T Q_{old}}{1 + V_{new}^T Q_{old} V_{new}}$$

where:

$$Q_{old} = (V_{old}V_{old}^T)^{-1}$$

6.1 Convergence of β_0

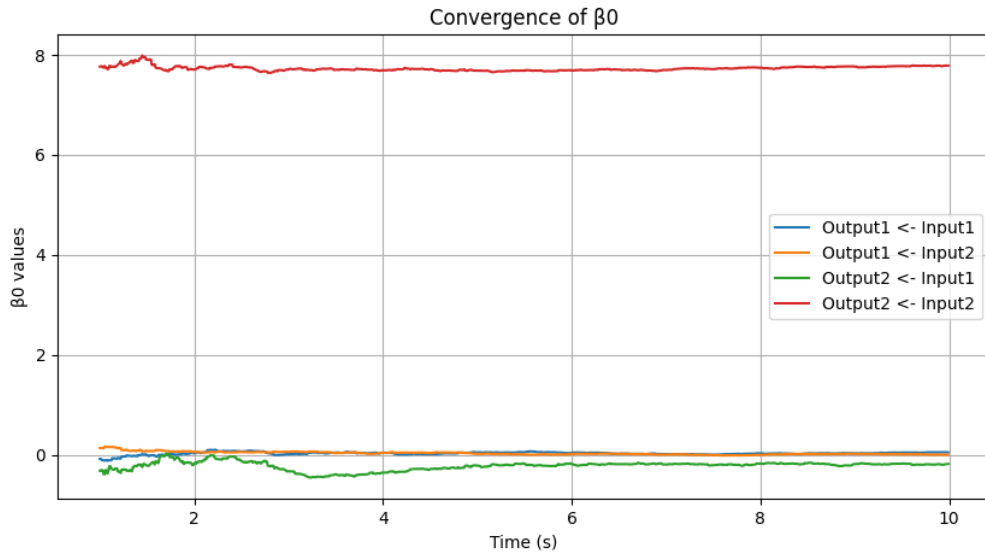


Figure 5: Convergence of coefficients of β_0 during recursive updating.

6.2 Convergence of α_1

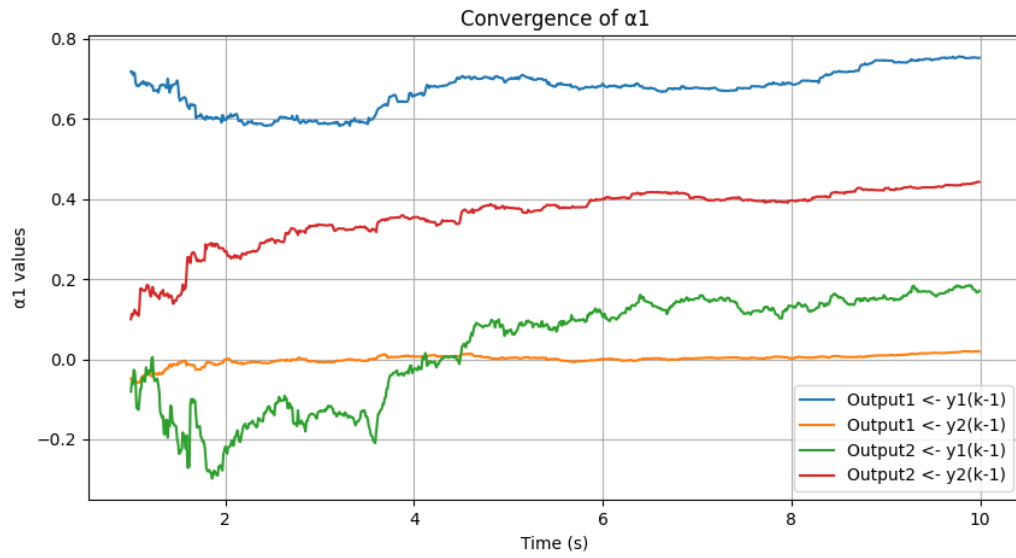


Figure 6: Convergence of coefficients of α_1 during recursive updating.

The parameters initially fluctuated due to limited data and gradually converged as more samples became available.

7. Conclusions

- ARX models successfully captured the dynamic behavior of the structural system.
- ARX(4) provided good reconstruction and prediction accuracy.
- BIC selected ARX(6) as the best balance between accuracy and complexity.
- Best pure prediction performance occurred near $p = 36$.
- Recursive identification demonstrated stable online convergence of model parameters.

Thus, ARX-based system identification proved effective for vibration-based structural monitoring.